

Pritam

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🐙 [GitHub](#)

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Java, SQL, VHDL
Technologies & Frameworks: Java Swing, AWT, OpenCV, Spring Boot
Machine Learning: Linear/Logistic Regression, Random Forests, Scikit-learn
Tools & Platforms: Git, Linux, IntelliJ IDEA, VS Code, Vim
Specializations: AI/ML, Robotics, PCB Designing

EDUCATION

Deenbandhu Chhotu Ram University of Science and Technology Sonipat, Haryana
Bachelor of Technology in Electronics & Communication Engineering Sep 2023 - Present
CBSE - 12 2022
Mathematics Grade: 96.2% New Delhi

EXPERIENCE

Embedded Application Intern Jul 2025 – Present
PCB CUPID, Bangalore, India Remote

- Built robots and embedded systems using Glyph development boards (C2, C6, H2) with focus on **precision control** and hardware-software integration for **production-ready applications**
- Designed and prototyped embedded solutions converting concepts to **scalable, practical systems** deployed in real-world environments
- Programmed microcontroller solutions achieving **seamless integration** between hardware components and software algorithms

Robotics Club Member and Mentor Oct 2023 – Present
THINKBOTS, DCRUST Murthal, Haryana

- Developed Line Follower and Multi-Terrain Gripper robots using Arduino and sensor arrays, achieving **95% accuracy** in path tracking
- Mentored **15+ junior members** on robotics fundamentals and Arduino programming, improving their project completion rate by **40%**
- Optimized robotics systems through collaborative troubleshooting, reducing debugging time by **30%** across team projects

PROJECTS

Forge - PCB Visualization CLI Tool — 🐙 [GitHub](#) — *Python, Click, SVG Processing, XML Parsing* Aug 2025

- Architected command-line tool for automated conversion of KiCad layer SVG files into seamless PCB visualizations, streamlining circuit board design workflow for engineers and hobbyists
- Engineered modular SVG processing pipeline integrating KiCad position files with coordinate transformation algorithms to generate comprehensive front and back PCB layout representations
- Implemented intelligent component placement system with geometric mapping algorithms, enabling precise positioning and orientation of component SVGs with sub-millimeter accuracy
- Developed robust CLI interface using Click framework with comprehensive error handling, progress tracking, and batch processing capabilities for production environments

Autonomous Line Following Robot — *Arduino Nano, C++, IR Sensors, PID Control* Mar 2025

- Engineered autonomous navigation system achieving 98% path accuracy using 8-IR sensor array with L298N motor driver, implementing real-time feedback control for competition-grade performance
- Designed and tuned PID control algorithm with adaptive parameter adjustment, enabling smooth navigation through 90° corners and dynamic track variations while minimizing overshoot and oscillation
- Optimized sensor calibration protocols and filtering algorithms, reducing false positive readings by 85% and ensuring stable operation across diverse lighting conditions and surface materials

- Implemented modular embedded architecture with configurable sensor thresholds, motor speed profiles, and debug interfaces for rapid prototyping and performance optimization

Dobosaurus - Chrome Dino Game Clone —  GitHub — *Java, Swing, Object-Oriented Design* Nov 2024

- Developed arcade-style platformer game featuring responsive jump mechanics, procedural obstacle generation, and real-time score tracking system with progressive difficulty scaling
- Implemented advanced graphics rendering pipeline with parallax scrolling background system, sprite animation frameworks, and collision detection using bounding box optimization for 60+ FPS performance
- Designed object-oriented architecture with separation of game logic, rendering, and input handling components, enabling extensible gameplay features and clean code maintainability

ACHIEVEMENTS & RECOGNITION

Smart India Hackathon — *Team Leader* 2024

- **2024:** Face Recognition Surveillance System and Communication Systems for Missing Persons or Items at Simhastha Ujjain for the Police Department

ADDITIONAL SKILLS

Operating Systems: Linux (Ubuntu, Arch Linux), Windows

Development Tools: Git, GitHub, Vim